

Build & Deploy Guide

This project includes a **C# backend** (OpenAPI-based), an **Angular frontend**, and uses **Ollama**, **Apache Tika**, and **PostgreSQL** for various services. The development and deployment environments are fully containerized using Docker.

Prerequisites

- [Docker](#)
- [Docker Compose](#)
- [GitHub CLI \(optional\)](#) (GitHub Workflows)

Getting Started Locally

1. Build & Start All Services

From the `/BitAndBeam` directory, run:

```
docker-compose up --build
```

This will:

- Build and run the C# backend
- Serve the Angular frontend
- Start Ollama Service
- Launch Apache Tika (document parsing)
- Set up a PostgreSQL database instance

2. Accessing Local Services

- Frontend: <http://localhost:8080>
- Backend API: <http://localhost:5001>
- PostgreSQL: localhost:5432 (credentials in `.env` or `docker-compose.yml`)
- Tika Server: <http://localhost:9998>
- Ollama: <http://localhost:11434>

GitHub Actions CI/CD Pipeline

This project includes a **GitHub Actions** pipeline for automated builds and deployments to a server.

What It Does

- Runs on every push to the `main` branch
- Builds Docker images for each service
- Pushes them to the GitHub Container Registry
- Remotely runs containers on the production server

Workflow File

See `.github/workflows/docker-ci.yml` for the CI/CD definition.